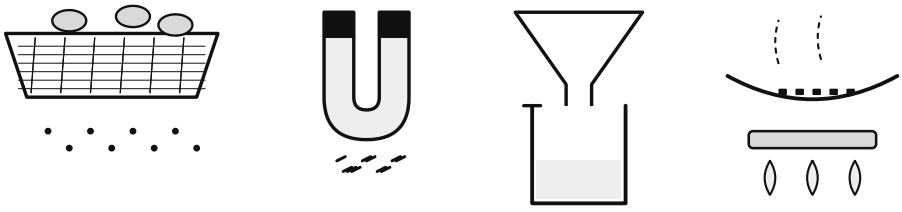
PHYSICAL AND CHEMICAL SCIENCE

Mixtures and Separation

What happens when substances mix, ways to separate mixtures, and recycling by melting and moulding



sieving **using a magnet** **filtering** **evaporating**

In this pack

Learn it: Mixing and separating ★☆☆
Fact check: Choose a method ★☆☆
Think it through: Separation puzzles ★★☆
Investigate: Which filter cleans muddy water best? ★★☆
Real-world case study: The SONO filter: clean water from a Bangladeshi invention ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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LEARN IT · C2 · MIXTURES

Mixing and separating

★☆☆ Grade 4

Name: Class: Date:

Learning goals

- Observe and describe what happens when different substances are mixed.
- Choose and use ways to separate mixtures.
- Explain how melting and freezing help us recycle materials.

In the kitchen, mixtures are everywhere: rice with tiny stones, dal soaking in water, sugar stirred into cha, spices ground together. Cooks are clever scientists. They know how to mix things and how to separate them again.

What happens when we mix?

When we mix substances, different things can happen. Look carefully and describe what you see.

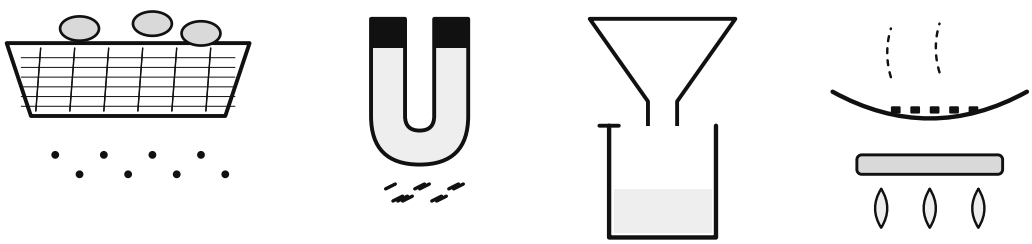
Mixture	What happens
sugar + water	the sugar **dissolves** and seems to disappear
sand + water	the sand does not dissolve; it **sinks** to the bottom
oil + water	they do not mix; the oil **floats** on top
vinegar + baking soda	it **fizzes**: a new gas is made
flour + water	a thick **paste** forms

⚠ Stay safe

Never taste or smell mixtures in science lessons unless your teacher says it is safe.

Ways to separate mixtures

We separate mixtures by using the different **properties** of the things in them: size, whether they dissolve, whether they are magnetic, or whether they float.



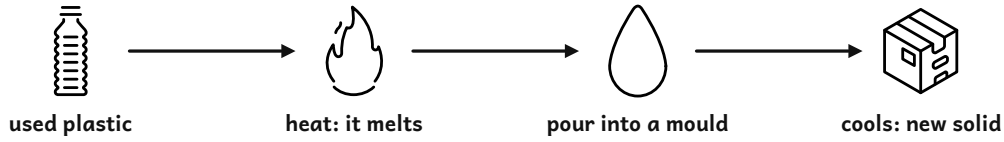
sieving **using a magnet** **filtering** **evaporating**

Four ways to separate a mixture.

- **Picking out**: removing stones from rice by hand.
- **Sieving**: a sieve lets small bits through and keeps big bits back (flour from lumps).
- **Winnowing**: tossing rice on a *kula* so the wind blows the light husks away.
- **Using a magnet**: pulling iron pins out of sand.
- **Pouring off** (decanting): letting mud sink, then pouring the water off the top.
- **Filtering**: pouring muddy water through filter paper or cloth; the water passes through, the mud stays behind.
- **Evaporating**: leaving salty water in the sun; the water evaporates and the salt is left.

Recycling with heat

Changes of state help us **recycle**. Plastic, glass and metals such as aluminium and iron can be **melted** by heating them, poured into a **mould**, and left to cool and **freeze** (become solid again) in a new shape. Old cans can become new pots; broken glass can become new bottles.



Melting and moulding to recycle.

Bangladesh connection

In Old Dhaka, small workshops melt scrap aluminium and pour it into sand moulds to make cooking pots (*hari* and *patil*). Collectors called *bhangari* buy old metal, paper and plastic from homes so they can be recycled.

Try this at home

Mix a spoon of sand, a spoon of salt and some water in a cup. Can you get the sand back? Can you get the salt back? (Hint: filter, then leave the water in the sun.)

Key words

mixture	two or more substances mixed but not joined
dissolve	break up and spread through a liquid
sieve	a mesh that separates big and small pieces
filter	separates bits of solid from a liquid
evaporate	a liquid turns into a gas
winnowing	using wind to blow light husks away from grain
mould	a hollow shape that liquid is poured into
recycle	make used materials into new things

FACT CHECK · C2 · MIXTURES

Choose a method

★☆☆
Grade 4

Name: Class: Date:

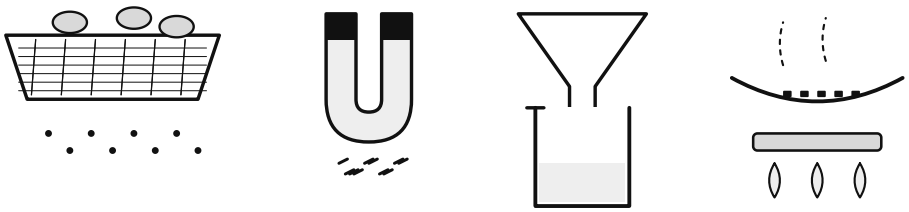
Learning goals

- Match mixtures to separation methods.
- Describe mixing.

1 Which method separates each mixture? [5]

Mixture	Method
stones in rice	
iron pins in sand	
sand in water	
salt in water	
husks and rice grains	

2 Name each method. [4]



A B C D

A	B	C	D

3 What happens? Write dissolves, sinks, floats or fizzes. [4]

Mixture	What happens
salt in water	
a stone in water	
oil in water	
vinegar and baking soda	

4 **Put the steps of recycling a metal can in order.** [3]

- ☐ take out the new pot
- ☐ collect old cans
- ☐ let it cool and become solid
- ☐ heat them until they melt
- ☐ pour the liquid metal into a mould

THINK IT THROUGH · C2 · MIXTURES

Separation puzzles

★★☆
Grade 4

Name: Class: Date:

Learning goals

- Plan how to separate a mixture.
- Explain using properties.

1 **Mina spilled a mixture of sand, salt, iron pins and dried peas. Plan how to get each one back. Explain each step.** [5]

2 **Why can't you separate salt from water with a sieve or a filter?** [2]

3 **Why must a mould be made of something with a much higher melting point than the metal poured into it?** [2]

4 **Is mixing vinegar and baking soda a good way to make something you can separate again? Explain.** [1]

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LEARN · PLAY · WONDER

Which filter cleans muddy water best?

★★☆ Grade 4

Name: Class: Date:

Learning goals

- Compare materials fairly.
- Use a rating scale.
- Draw conclusions.

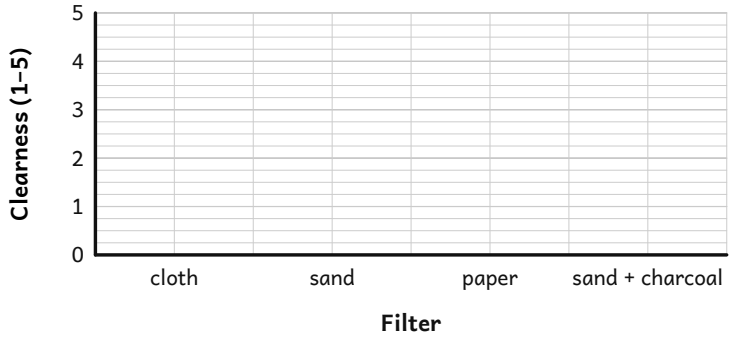
Class 4C poured 100 ml of the same muddy water through four different filters in funnels. They rated how clear the water was from **1** (still very muddy) to **5** (very clear).

1 **Complete the plan.** [3]

Change	
Measure	
Keep the same	

2 **Draw a bar chart.** [3]

Filter	cotton cloth	sand	filter paper	sand + charcoal
Clearness (1–5)	2	3	4	5



3 **Which filter was best? Suggest why.** [2]

.....

.....

4 **The water from the best filter looks clear. Is it safe to drink? Explain.** [2]

.....

.....

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LEARN · PLAY · WONDER

The SONO filter: clean water from a Bangladeshi invention

★★★ Grade 4

Name: **Class:** **Date:**

Learning goals

- Apply filtering to a real problem.
- Understand how a scientist designed a solution.
- Evaluate the solution.

Science · Health · Design · Economics

Millions of tube-wells in Bangladesh give water that is free of the germs found in ponds. But scientists discovered that the water from many wells contains **arsenic**, a poison that dissolves into the water from rocks deep underground. You cannot see, smell or taste it, and drinking it for years can cause serious illness.

Dr Abul Hussam, a chemist born in Kushtia, designed a simple filter called **SONO**. Water trickles through buckets filled with layers of **sand**, **brick chips**, **charcoal** and a special **iron mixture** that grabs the arsenic. It needs no electricity. In 2007 his invention won a major international prize for engineering, and many thousands of families have used it.

Based on information about arsenic in groundwater in Bangladesh and the SONO filter.

1 **Match each filter layer to its job.** [3]

Draw a line to match each pair, or write the letter.

sand and brick chips ●	● **A** grabs the dissolved arsenic
iron mixture ●	● **B** removes smells and tiny bits
charcoal ●	● **C** trap bits of mud and sand

2 **Why can't ordinary cloth filter remove arsenic?** [2]

.....

.....

3 **Give two reasons why the SONO filter is a good design for villages.** [2]

.....

.....

4 **What might each person think?** [3]

Person	View
a mother	
a scientist	
a shopkeeper selling filters	

Show what you know

★★★
Grade 4

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

M	M	O	J	S	I	I	F	V	S
D	E	V	P	A	C	S	I	L	I
I	O	L	V	L	M	P	L	A	E
S	O	L	T	T	N	D	T	H	V
S	M	I	X	T	U	R	E	M	E
O	S	T	N	W	W	I	R	A	L
L	A	O	K	O	H	L	V	G	E
V	N	A	S	V	R	I	K	N	W
E	D	V	L	W	P	C	L	E	D
M	O	U	L	D	E	Y	N	T	N

dissolve filter magnet melt mixture mould salt sand sieve

2 Which method gets salt back from salty water? [1]

- ◯ sieving
- ◯ filtering
- ◯ evaporating
- ◯ using a magnet

3 A farmer tosses rice on a *kula* in the wind. This is called... [1]

- ◯ winnowing
- ◯ filtering
- ◯ melting
- ◯ dissolving

4 Draw and label equipment to filter muddy water. [3]

Filtering

5 **Challenge. Explain why recycling a metal pot by melting it is a reversible change, but burning paper is not.** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
describe what happens when substances are mixed	○	○	○
choose a method to separate a mixture	○	○	○
plan how to separate a mixture of four things	○	○	○
test which filter works best	○	○	○
explain recycling by melting and moulding	○	○	○
explain how the SONO filter works	○	○	○